Survival from sports-related sudden cardiac arrest: In sports facilities versus outside of sports facilities

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Background We sought to evaluate frequency, characteristics, and outcomes of sudden cardiac arrest (SCA) during sports activities according to the location of occurrence (in sports facilities vs those occurring outside of sports facilities).

Methods and results This is an observational 5-year prospective national French survey of subjects 10 to 75 years old presenting with SCA during sports (2005-2010), in 60 French administrative regions (covering a population of 35 million people). Of the 820 SCA during sports, 426 SCAs (52%) occurred in sports facilities. Overall, a substantially higher survival rate at hospital discharge was observed among SCA in sports facilities (22.8%, 95% CI 18.8-26.8) compared to those occurring outside (8.0%, 95% CI 5.3-10.7) ($P < .0001$). Patients with SCA in sports facilities were younger (42.1 vs 51.3 years, $P < .0001$) and less frequently had known cardiovascular diseases ($P < .0001$). The events were more often witnessed (99.8% vs 84.9%, 0.0001), and bystander cardiopulmonary resuscitation was more frequently initiated (35.4% vs 25.9%, $P = .003$). Delays of intervention were significantly shorter when SCA occurred in sports facilities (9.3 vs 13.6, P=0.03), and the proportion of initially shockable rhythm was higher (58.8% vs 33.1%, $P < .0001$). Better survival in sports facilities was mainly explained by concomitant circumstances of occurrence (adjusted odds ratio 1.48, 95% CI 0.88-2.49, $P = .134$).

Conclusions Sports-related SCA is not a homogeneous entity. The 3-fold higher survival rate reported among sports-related SCA is mainly due to cases that occur in sports facilities, whereas SCA during sports occurring outside of sports facilities has the usual very low rate of survival. (Am Heart J 2015;170:339-345.e1.)

Sudden cardiac arrest (SCA) during sports activities always attracts considerable media attention. To date, the problem of SCA during sports has been mainly viewed through the prism of young competitive athletes,[1-6] and little data have been available from the general population.

We have recently reported the results of a large prospective 5-year French national program aiming to document the burden and outcomes of sports-related SCA from a general population perspective.[7] Our results, recently supported by data from the Netherlands,[11] have emphasized that, overall, sports-related SCA showed relatively good outcomes, compared to SCA not associated with sports activities. This better survival observed among sports-related SCAs may be the result of specific intrinsic characteristics of sports participants (vs nonsports participants) and also from significant differences in the circumstances of occurences, notably the actions initiated by bystanders. To better understand this better outcome among sports-related SCA, we hypothesized that sports–related SCA was not a homogeneous group and that the better survival during sports activities mainly reflects improved outcomes in SCA in sports facilities.

In the present article, we thus sought to compare characteristics and outcomes of sports-related SCA between occurrences in sports facilities versus those occurring outside of sports facilities.

Materials and methods

We performed an ad hoc analysis of a prospective observational 5-year national French survey of subjects 10 to 75 years old presenting with SCA during sports. The study design and population have been described previously.[7-10] Briefly, this 5-year prospective study was carried

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out according to Utstein guidelines, in France between April 2005 and April 2010. The registry is a prospective national database offering a global view of SCA during sports, including cases from recreational as well as competitive sports. We obtained close collaboration from the French system of mobile intensive care units (Service d'Aide Médicale Urgente). All of the 96 administrative districts throughout France were invited to participate, and 60 of these districts eventually participated (which represent approximately 70% [33,948,400] of the French population aged 10-75 years), including overall 121 hospitals and intensive care units (see online Appendix for participating centers). The objective was to describe SCA during sports in terms of incidence, subject characteristics, emergency care, and outcomes at admission and hospital discharge (including survival as well as functional outcome).

Sudden cardiac arrrest in the setting of sports activities was defined as cardiac arrest occurring during or within 1 hour of cessation of sports activity.[7,11-13] Sports-related SCAs related to trauma (other than for cases of commotio cordis) were excluded. Subjects aged between 10 and 75 years were included in the study because the context of physical activity can be difficult to assess in the very young or in those older than 75 years (who rarely participate in sports). For each subject, the site of occurrence was collected and classified as in sports facilities or cases occurring outside of sports facilities. Sports facilities were defined as activities practiced (i) in places specially designed, regardless of their size (including stadium, tennis facilities, etc), or (ii) in the context of competition. By contrast, setting was defined as outside of sports facilities for others—individuals who performed sports activities without any structure or organized competition.

We obtained excellent cooperation from the French system of Emergency Medical Services (in France, Service d'Aide Médicale Urgente) as well as from different intensive care units throughout the 60 participating districts. Because of the expected difficulty in case ascertainment through this community-based approach during the study period, we used 2 complementary independent methods to optimize case detection: (i) an approach based on prospective case reporting system from different emergency medical services (EMS) units, which systematically attended cases of cardiac arrest or sudden collapse, and (ii) a media search for cases, using a Web-based monitoring via a Web-based search engine (Google Reader), carried out continuously by a specific working group checking on a daily basis. All press releases not reported by EMS prompted further contact with the local EMS and/or local hospital intensive care unit teams for collection of detailed data on resuscitation provided in the field, hospital care, and survival status at hospital discharge.

In France, a unique EMS exists—Service d'Aide Medicale Urgente—which is a special unit within a hospital designed to receive and dispatch urgent medical calls. Paramedics are dispatched and usually arrive first on the scene to initiate chest compression and ventilation and rhythm assessment as well as prompt defibrillation (when appropriate). Then, a medical team is systematically dispatched in an ambulance, including a physician specialized in emergency, and initiates advanced life support, including intravenous lines, mechanical ventilation, and inotropic support. Patients in whom restoration of spontaneous circulation is achieved are then referred to a tertiary center with intensive care unit and coronary intervention facilities available 24 hours a day, 7 days a week.[14,15] In France, the implementation and organization of the EMS are homogeneously distributed throughout the country and provide care via common guidelines/protocols across districts.[14]

Collected variables

Variables collected for each case report file included demographic information and medical history (known heart disease, cardiovascular risk factors, and cardiac symptoms during the preceding days), data regarding sports activities (type of sports, collective or individual practice, leisure or competitive event, and level of exercise estimated at the time of collapse), and circumstances of collapse (presence of one/multiple witnesses and cardiopulmonary resuscitation (CPR) initiation, initial cardiac rhythm recorded, bystander use of automatic external defibrillator, and defibrillator shocks delivered). For each subject, site of occurrence was collected and classified as sports facilities (including sports ground, stadium, and fitness center) or cases occurring outside of such facilities. Level of exertion was classified as light, moderate, or high, by ambulance officers and/or hospital medical staff according to metabolic equivalents and a prespecified scale.[13] In addition, information regarding survival status at hospital admission was all collected using information from the EMS. Data regarding survival status at hospital discharge were recorded prospectively by the INSERM U970. For survivors, neurologic status at discharge was independently evaluated at discharge by the physician in intensive care unit or cardiology department, using the Cerebral Performance Categories (CPC) score; a lower CPC score (1-2) indicates a better neurologic outcome.[16] Files were prospectively and regularly reviewed (every 6 months) by an independent events committee.

Statistical analysis

This report was prepared in compliance with the STROBE checklist for observational studies.[17] Characteristics of SCA occurring during or immediately after sports were described as mean ± SD, proportions, median, and interquartile ranges (delays), as appropriate. Percentages were calculated based on the total number of known events. 95% CIs for proportion were calculated. Qualitative variables were compared using the χ^2 and Fisher exact tests, whereas quantitative variables were compared using Mann-Whitney *U* test, analysis of variance, or Kruskal-Wallis test for multiple comparisons. In addition to age and sex, all covariates that reached a significance level of $P < .15$ for survival (including time of intervention—call to EMS arrival) were then included in an initial multivariable regression model.[18] Odds ratios (ORs) for survival to hospital discharge and their 95% CIs were calculated. All tests were 2 tailed, and $P < .05$ was considered to indicate statistical significance. All data were analyzed at INSERM, Unit 970, Cardiovascular Epidemiology and Sudden Death, Paris, using STATA software version 11.0. (Lakeway Drive, TX).

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Results

Among the 820 patients included in the study, 426 (52%) presented with SCA in sports facilities, and 394 (48%) occurred outside of such sports facilities. A simplified Utstein reporting template for the 820 cases, according to the location of occurrence, is represented Figure 1. Among cases occurring in sports facilities, 186 (43.7%) took place in a stadium; 117 (27.4%), in a fitness center or a gymnasium; and 123 (28.9%), in other specific sports areas, including swimming pools and tennis courts (in 31 and 15 cases, respectively). During 5 years of observation, a total of 820 sports-related SCA were recorded: 418 (51%) first reported by EMS and 402 (49%) from press reports. Sports-related SCA occurred actually during sports activity in 92%, immediately after (within 30 minutes) in 7.4%, and only rarely after 30 minutes from the cessation of sports activity.

Baseline characteristics of the population according to the site of occurrence are shown in Table I. Patients presenting with SCA in a sports facilities were younger (42.1 vs 51.3 years, $P < .0001$) and less frequently had known heart disease and/or >1 cardiovascular risk factor (5.2% vs 18.5%, $P < .0001$). Among the cases aged ≤35 years old, 152 (80.8%) occurred in the setting of sports facilities. Among the subgroup of sports-related SCA occurring among young competitive athletes, 49 (98.0%) occurred in sports facilities. Compared to SCA occurring outside of sports facilities, those in sports facilities occurred more often in the setting of team sports activities (50.2% vs 15.0%, $P < .0001$) as well as during a higher level of exercise ($P < .0001$).

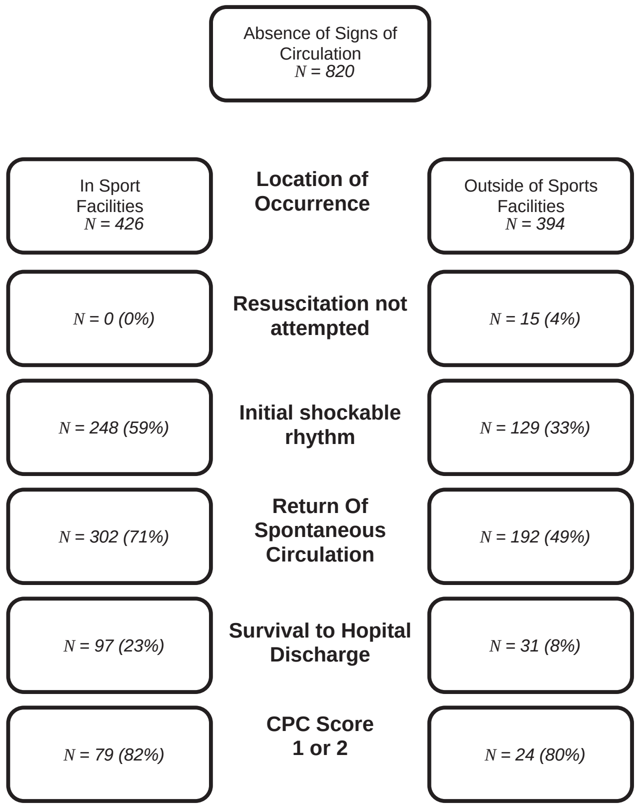
Characteristics of the SCA event according to their site of occurrence are listed in Table II. Sudden cardiac arrest in sports facilities was more often witnessed (99.8% vs 84.9%, $P < 0.0001$), and CPR was significantly more often initiated before EMS arrival (35.4% vs 25.9%, $P = .003$). Use of automatic external defibrillators (AED) by the public, before EMS arrival, was extremely rare (<1% overall), with no significant difference according to the site of occurrence ($P = .62$). Intervention delays significantly differed according to the location of occurrence: call to EMS arrival was 7.4 minutes (5.1-11.1) in sports facilities versus 11.1 minutes (7.9-14.8) in those that occurred outside of such facilities ($P = .03$); call to first defibrillator shock delay was 9.3 minutes (6.4-11.5) and 13.6 minutes (10.3-15.6), respectively ($P = .03$). Initial rhythm differed significantly according to the site of SCA, with a higher proportion of shockable ventricular arrhythmias (ventricular tachycardia and fibrillation) for SCA in sports facilities (58.8% vs 33.1%, $P < .0001$).

Overall, survival at hospital discharge was much higher when SCA occurred in sports facilities (22.8%, 95% CI 18.8-26.8) compared to those occurring outside (8.0%, 95% CI 5.3-10.7) ($P < .0001$). Overall, AED was used in 494 cases. However, among those, we denoted only 5 cases of defibrillation by public witnesses (not related to EMS), before EMS arrival. Rate of return of spontaneous circulation was higher among patients experiencing SCA in sports facilities (71.0% vs 50.5%, $P < .0001$). Similarly, patients with SCA in sports facilities had higher survival rates at hospital admission (40.6% vs 20.3%, $P < .0001$) and during hospital stay (56.1% vs 38.8%, $P < .0001$). The proportion of survivors with favorable neurologic outcome (CPC 1 or 2) was similar between the 2 groups, 82.3% and 80.0% in sports facilities and outside of facilities, respectively ($P = .77$) (Figure 2).

The crude association between sports facilities and survival was highly significant (OR 2.83, 95% CI 1.88-4.25, $P < .0001$). After adjusting for subject characteristics (age, gender, and history of heart disease), location of occurrence (in sports facilities vs outside of sports facilities) was attenuated but still significantly associated with survival (OR 2.00, 95%CI 1.26-3.19, $P = .003$). After adjusting for further other variables related to circumstances of occurrences (presence of witnesses, bystander CPR, delay of interventions, and initial rhythm at EMS arrival), location of occurrence was no longer significantly associated with survival (OR 1.48, 95% CI 0.88-2.49, $P = .134$), whereas bystander CPR (OR 3.73, 95% CI 2.19-6.39, $P < .0001$), initial shockable cardiac rhythm (OR 3.71, 95% CI 2.07-6.64, $P < .0001$), and time from collapse to start of CPR (OR 1.32, 95% CI 1.08-1.61, $P = .006$) remained significantly associated with survival at hospital discharge.

Of the 203 subjects where cause of death was ascertained, structural heart disease was present in 189, principally including coronary artery disease (93.1%), whereas channelopathy or idiopathic ventricular fibrillation was only detected in 14 cases (6.9%). No significant difference was found in comparing cases from sports facilities and nonsports facilities.

Figure 1



Utstein reporting template for the 820 SCA according to the location of occurrence.

Discussion

From this 5-year prospective nationwide evaluation of SCA during sports, we have firstly demonstrated that approximately half of sports-related SCA took place in the setting of sports facilities, in France. We then emphasized the extent to which these 2 different settings (in sports facilities vs outside of sports facilities) were associated with different outcomes. Finally, our results suggest that the 3-fold higher survival in sports facilities is mainly the result of patients' characteristics as well as specific circumstances of occurrence (notably action of bystanders).

We have previously reported the overall estimate of sports-related SCA in France between 4.6 and 16.7 cases per million per year,[7] giving the opportunity to estimate that sports-related SCAs represent a relatively small proportion of overall SCA (approximately 1%-3%), a fact that is likely to encourage sports activities. In the general population, the site of occurrence of SCA (at home vs in public places) is associated with major differences in subject characteristics and survival rates.[19-22] It has been well demonstrated that documenting such differences is of major importance for optimizing preventive strategies, including AED dissemination and the evaluation of their cost-effectiveness.[19] Regarding sports-related SCA, the proportion of SCA, which occurred in the setting of sports facilities, has not been previously evaluated. Sports practices may differ from one area to another, with French participants possibly more likely to exercise outside of sports facilities.[23] Thus, the ratio of 50% of sports-related SCA in sports facilities may represent an underestimation compared to other industrialized countries, such as the United States.

We recently reported an overall survival rate of almost 16% at discharge after community-based sports-related SCA overall.[7] This survival rate is much higher than usually described in situations not related to sports, approximately 8% in a recent meta-analysis.[24] The recent data from Dutch province of North Holland also demonstrated that persons suffering SCA during physical exercise were more likely to survive (46% vs 17% in non–exercise-related SCA).[11] Our results allow better understanding of the reasons for the benefit in survival of sports-related events—that this relatively higher rate of survival after SCA during sports compared to overall SCA is mainly the result of better outcomes from SCA in sports facilities, whereas sports–related SCAs occurring outside such facilities have a similar prognosis to that of the general population.

In our study, the better survival observed after SCA in sports facilities appears to be mainly explained by not

Table I. Subject characteristics and circumstances of occurrence of sudden cardiac death during sports according to site of occurrence

	In sports facilities	Outside of sports facilities	*P*
No. of collected cases	426	394	
Demographic data			
Age (y)			
Mean	42.1 ± 14	51.3 ± 13	<.0001
Male sex, n (%)	404 (94.8)	373 (94.7)	.895
Young competitive athlete*	49 (11.5)	1 (2.5)	<.0001
History of known heart disease/ >1 CVRF†	22 (5.2)	72 (18.5)	<.0001
Time of occurrence			
Season			<.0001
Winter	153 (17.6)	83 (21.0)	
Spring	147 (34.5)	11 (28.2)	
Summer	95 (22.3)	148 (37.6)	
Autumn	109 (25.6)	52 (13.3)	
Week			
Weekend	186 (43.7)	150 (38.1)	.078
Day			.005
Morning (6-12 h)	265 (34.1)	22 (51.2)	
Afternoon (12-18 h)	340 (43.7)	19 (44.2)	
Evening (18-24 h)	172 (22.1)	2 (4.6)	
Sport setting			
Sport activities			<.0001
Team sports	214 (50.2)	59 (15.0)	
Individual sports	212 (49.8)	335 (85.0)	
Intensity of exercise			<.0001
Light	14 (3.3)	18 (4.6)	
Moderate	203 (48.1)	255 (65.4)	
Vigorous	205 (48.6)	117 (30.0)	

Data are expressed as number (percentages), unless otherwise specified. Data were missing on history of known heart disease (for 8 subjects), cardiovascular risk factors (11), and intensity of exercise (8). Percentages were calculated based on the total number of known events.

* A "young competitive athlete" was defined as any person, aged 10 to 35 years, who participated in an organized sports program (team or individual sports) requiring regular competition and training according to the National Registry from the Ministère de la Santé et des Sports (individuals participating in college-sponsored intramural sports were not considered as young competitive athletes).

† Cardiovascular risk factor included known/treated diabetes mellitus, dyslipidemia, systemic hypertension, obesity, current smoker, and family history of premature coronary heart disease.

Table II. Management of sudden cardiac death and mortality rates according to site of occurrence

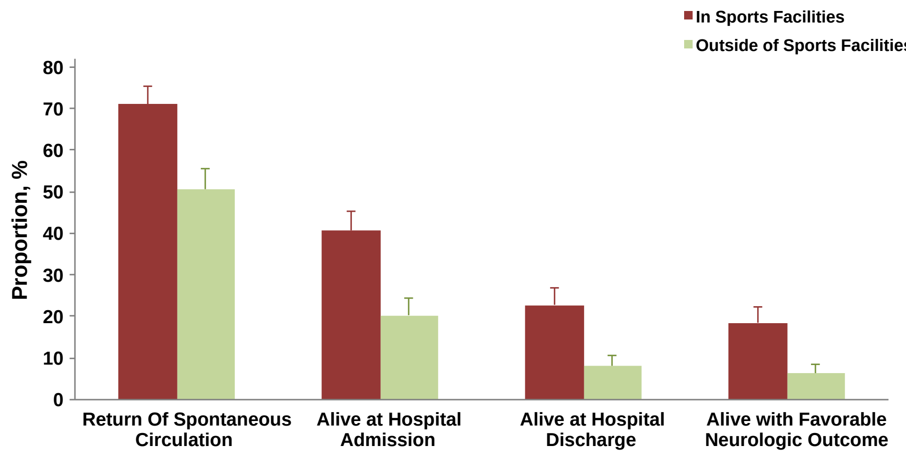
	In sports facilities, n = 426	Outside of sports facilities, n = 394	*P*
Sudden death witnessed	425 (99.8)	331 (84.9)	<.0001
Bystander CPR	151(35.4)	101 (25.9)	.003
Public AED use	4 (0.9)	1 (0.3)	.62
First monitored rhythm			<.0001
Ventricular fibrillation/ tachycardia	248 (58.8)	129 (33.1)	
Asystole	127 (30.1)	220 (56.4)	
Pulseless electrical activity	47 (11.1)	40 (10.3)	
Sinus rhythm	0 (0)	1 (0.3)	
Shock by external defibrillator	309 (72.5)	185 (47.0)	<.0001
Delay (min), median (IQR)			
Call to EMS arrival	7.4 (5.1-11.1)	11.1 (7.9-14.8)	.03
Call to first defibrillator shock	9.3 (6.4-11.5)	13.6 (10.3-15.6)	.03

Data are expressed as number (percentages), unless otherwise specified. Abbreviation: *IQR*, interquartile range. Data were missing on presence of witness (4) and monitored rhythm (12). Percentages were calculated based on the total number of known events.

only associated factors (presence of witnesses, CPR, and shorter delay) but also subject characteristics more conducive to survival,[25] and thus, occurrence in sports facilities was not independently associated with survival. Subject characteristics of SCA in sports facilities varied markedly compared with those who "died" outside of sports facilities, with subjects almost 10 years younger, with no known heart disease in most cases (>95%). However, those individuals' differences appear to explain only a relative part of the higher survival in sports setting. On the other hand, the circumstances of occurrence also differ, with SCAs in sports facilities more often being witnessed and witnesses being more likely to initiate CPR before EMS arrival. Accordingly, initial rhythm differed between groups, with a higher frequency of shockable rhythm in patients who presented SCA in a sports facilities, supporting the potential benefit of CPR to maintain shockable rhythm as well as emphasizing the importance of delay to intervention.[26-28] The 60% figure of shockable rhythm in sports facilities is higher than has been reported for general emergency rescue responders in recent years and even for some inhospital data, and this important point underlines the need for systematic availability of AEDs in the setting of public sports facilities.

Our results, highlighting the high rate of witnessed events with shockable rhythm in sports facilities, emphasize the urgent need to focus on the very early period after SCA, which includes CPR initiation by a bystander, and support a campaign of information regarding the need for a systematic availability of AEDs in sports settings.[29-31] In this regard, the recently published experience from in Seattle and King County from 1996 to 2008, showing that survival to hospital discharge may reach 54%, emphasizes the extent to which population information to basic life support as well as larger AED dissemination may improve outcomes after SCA in sports facilities.[29] The very low AED use by public bystanders in our study appears to be the result, at least in part, of limited AED availability (as AED availability in stadium or other sports facilities is not yet mandatory in France, with no clear recommendations, in contrast to the case of the United States).[32] In France, legalization of their use by public was only recently enacted (May 5, 2007).[33] There may also be a reluctance to use AEDs by the public. Indeed, we noted many cases where an AED was present on site but was not used by the witnesses (despite online telephone recommendation), underlying that beyond AED availability, other issues need to be addressed. Despite this very low frequency of AED use prior arrival of EMS (in only 5 cases) and given that no more than one-third of bystanders initiated CPR, survival after SCA in sports facilities was notably high, 22% at hospital

Figure 2



Outcomes after SCA according to the location of occurrence (in sports facilities vs outside of sports facilities).

discharge. Therefore, there is still a room for improvement, with the posibility of even better survival through widespread use of AED and better public awareness in CPR, especially through education programs. Our current findings should encourage public authorities as well as educators to not only extend the number of AED placed in such settings but also provide simple instructions on AED use, during regular training sessions on resuscitation. According to the known key role of CPR initiation as well as early defibrillation, our data emphasized the substantial room for improvement of public knowledge and attitudes toward basic life support.

Sports facilities have been identified as being particularly suitable for AED deployment.[34,35] Such programs are now increasing dramatically across countries; however, their cost-effectiveness has not been carefully addressed so far. This is mainly because there is a lack of data on sports-related SCA from the community. Specific guidelines are now available regarding cardiovascular safety in sports facilities.[35] It is of particular importance to provide characterization of SCA occurring in such areas to guide public health policies. The lack of information regarding incidence, subject characteristics, and circumstances of SCA during sports in various locations has previously limited comprehensive assessment of AED effectiveness, including cost-effectiveness.[36] However, our data, as well as recent findings from United States,[37] underline the need for systematic availability of AEDs in the setting of public sports facilities.

Although our study represents the first description of sports-related SCA in the community examined by location of occurrence, we acknowledge some limitations. First, this study was observational, and therefore, we were only able to demonstrate associations with survival without establishing a true causal link. In addition, there are other potential confounding factors that we may not have considered or measured, such as type or duration of sports activities. Second, although some resuscitation variables were not systematically collected, the risk for having a differential bias (in sports facilities vs outside of sports facilities) is limited, due to the homogeneous nature of the EMS and hospital protocols for such cases, across France. Third, patterns and locations of exercise as well as the response to events may be different in France compared to other countries, and this may limit the generalizability of our findings. Fourth, although the SCA before exercise can be of interest because there are some autonomic changes before competitive sports and such events occur in athletic facilities, we may have lack some cases according to the definition of sports-related SCA. Finally, most of sports-related SCA are witnessed (92% in the whole cohort), and we cannot rule out a differential bias regarding the report of unwitnessed SCA, occurring mostly outside of sports facilities.

In conclusion, sports-related SCA is not a homogeneous entity and covers 2 different settings, with major differences according to whether they occurred in sports facilities or elsewhere. The substantially higher survival rate reported among sports-related SCA is mainly due to cases that occur in sports facilities, whereas SCA during sports occurring outside of sports facilities has the usual very low rate of survival.

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Disclosure

None declared.
All authors have approved the final article.

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Appendix. Participating centers

The investigators and coordinators participating in the study were as follows:

District #02 Aisne, Dr Fréderic Degrootte, MD; District #05 Hautes-Alpes, Dr Michel Tashan, MD; District #06 Côte d'Azur, Dr Dominique Grimaud, MD; District #13 Bouches–du-Rhône, Dr Véronique Vig, MD; District #14 Calvados, Dr Daniel Bonnieux, MD; District #16 Charente, Dr Rémy Loyant, MD; District #17 Charente-Maritime, Dr Francis Mounios, MD; District #18 Cher, Dr François Bandaly, MD; District #21 Côte-d'Or, Dr Marc Freysz, MD; District #22 Côtes-d'Armor, Dr Christian Hamon, MD; District #24 Dordogne, Dr Michel Gautron, MD; District #25 Doubs, Dr Marie-Céline Maillot, MD; District #26 Drôme, Dr Claude Zamour, MD; District #28 Eure-et-Loir, Dr Nicolas Letellier, MD; District #29 Finistère, Dr Claude Lazou, MD; District #31 Haute-Garonne, Dr Jean-Louis Ducasse, MD; District #32 Gers, Dr Jean-Maurice Guez, MD; District #33 Gironde, Dr Michel Thicoipe, MD; District #34 Hérault, Dr Pierre Benatia, MD; District #36 Indre, Dr Louis Soulat, MD; District #37 Indre-et-Loire, Dr Jacques Fusciardi, MD; District #38 Isère, Dr Katell Berthelot, MD; District #39 Jura, Dr Antoine Elisseeff, MD; District #40 Landes, Dr Rachel Ricard, MD; District #41 Loir-et-Cher Dr Solo Randriamalala, MD; District #42 Loire, Dr Thomas Guerin, MD; District #44 Loire Atlantique, Dr Valerie Debierre, MD; District #45 Loiret, Dr Sophie Narcisse, MD; District #48 Lozère, Dr Marc Chassing, MD; District #51 Marne, Dr Alain Léon, MD; District #52 Haute-Marne, Dr Jacques Milleron, MD; District #55 Meuse, Dr Michel Vedel, MD; District #56 Morbihan, Dr Françoise Charland, MD; District #59 Nord, Dr Nordine Benameur, MD; District #60 Oise, Dr Thierry Ramaherison, MD; District #62 Pas-de-Calais, Dr Luce Hapka, MD; District #63 Puy-de-Dôme, Dr Jacques Meyrieux, MD; District #64 Pyrénées-Atlantiques, Dr Isabelle Pouyanne, MD; District #65 Hautes-Pyrénées, Dr Jérôme Khazaka, MD; District #67 Bas Rhin, Dr Jean-Claude Bartier, MD; District #68 Haut-Rhin, Dr Bruno Goulesqueb, MD; District #69 Rhône, Dr Pierre-Yves Gueugniaud, MD; District #70 Haute-Sâone, Dr Toufiq El Cadi, MD; District #71 Sâone-et-Loire, Dr Bruno Girardet, MD; District #72 Sarthe, Dr Christophe Savio, MD; District #75 Paris, Dr Daniel Jannière, MD; District #76 Seine-Maritime, Dr Bertrand Dureuil, MD; District #77 Seine et Marne, Dr Jean-Yves Le Tarnec, MD; District #78 Yvelines, Dr Clotilde Cazenave, MD; District #80 Somme, Dr Christine Ammirati, MD; District #81 Tarn, Dr Marie-Gabrielle Vaissière, MD; District #83 Var, Dr Jean-Jacques Raymond, MD; District #84 Vaucluse, Dr Philippe Olivier, MD; District #88 Vosges, Dr Hubert Tonnelier, MD; District #89 Yonne, Dr Monique Duche, MD; District #90 Territoire-de-Belfort, Dr Adel Kara, MD; District #92 Hauts-de-Seine, Dr Michel Baer, MD; District #93 Seine-Saint-Denis, Dr Fréderic Adnet, MD; District #94 Val-de-Marne, Dr Chantal Vallier, MD; District #95 Val-d'Oise, Dr Cédric Ramaut, MD.